

OTT Customer Churn Analysis

Functional Requirements Document (FRD)

Prepared By: Abhijeet Kedar

Purpose

This document defines the functional requirements, system behavior, dashboard specifications, and reporting capabilities for the OTT Customer Churn Analysis solution.

System Overview

The solution uses Excel for data preparation, SQL Server for business analysis, and Power BI for dashboard development and reporting.

Data Source

- Customer
- Subscription
- Usage
- Support
- Churn

Functional Requirements

ID	Functional Requirement
FR-01	Import customer datasets into the solution.
FR-02	Validate and clean data before analysis.
FR-03	Create relationships between all tables using CustomerID.
FR-04	Generate calculated columns and DAX measures.
FR-05	Calculate business KPIs.
FR-06	Execute SQL queries for business analysis.
FR-07	Build interactive Power BI dashboards.
FR-08	Support filtering, drill-through, and tooltips.
FR-09	Export dashboards and reports.

Dashboard Requirements

- **KPIs:**
 - Total Customers
 - Active Customers
 - Churned Customers
 - Revenue
 - Churn Rate
- **Charts:**
 - Churn by Engagement
 - Revenue by Subscription
 - Support vs Churn
 - Payment vs Churn
 - Customer by City
 - Churn by Age Band
- **Slicers:**
 - Gender
 - City
 - Subscription
 - Payment Method
 - Engagement Level

Business Rules

- CustomerID must uniquely identify each customer.
- Churn Status must contain only Active or Churned.
- Upgrade Accepted cannot be TRUE when Upgrade Offered is FALSE.
- Revenue must be greater than or equal to zero.

DAX Measures

Measure	Purpose
Churn Rate	Calculates churn percentage
Upgrade Acceptance Rate	Calculates accepted upgrades
ARPU	Average revenue per customer
Total Revenue	Total customer revenue
Average Rating	Customer satisfaction
Active Customers	Count of active customers

Testing & Validation

- Verify KPI calculations.
- Verify slicer functionality.
- Verify drill-through functionality.
- Verify tooltip functionality.
- Validate SQL results with dashboard values.

Non-Functional Requirements

- Dashboard response time should be under 5 seconds.
- Dashboard should support interactive filtering.
- Reports should refresh without errors.
- Visuals should maintain consistent formatting.
- Dashboard should be easy to use.

Dashboard Pages

Page 1 - Executive Overview

Purpose

Provides a high-level summary of platform performance by displaying key business KPIs such as Churn Rate, Upgrade Acceptance Rate, Total Revenue, Average Watch Hours, and Total Support Tickets. This page helps stakeholders quickly assess overall business performance.

Page 2 - Churn Investigation

Purpose

Analyzes customer churn by comparing engagement levels and support tickets. The page includes drill-through functionality for customer-level details and tooltips displaying Tenure, Revenue, and Support Tickets to support detailed churn analysis.

Page 3 - Upgrade Behavior

Purpose

Displays upgrade offers, accepted upgrades, and upgrade acceptance rate. It also analyzes customer engagement across different subscription plans to identify customers with the highest upgrade potential.

Page 4 - Revenue & Value

Purpose

Provides revenue analysis by Subscription Type and City while highlighting top revenue-generating customers. This page supports profitability analysis and customer value assessment.

Page 5 - Support & Experience

Purpose

Examines customer support activity and satisfaction by comparing Support Tickets, Customer Rating, and Ticket Categories. It helps identify customers who may be at higher risk of churn due to poor support experiences.

Dashboard Design

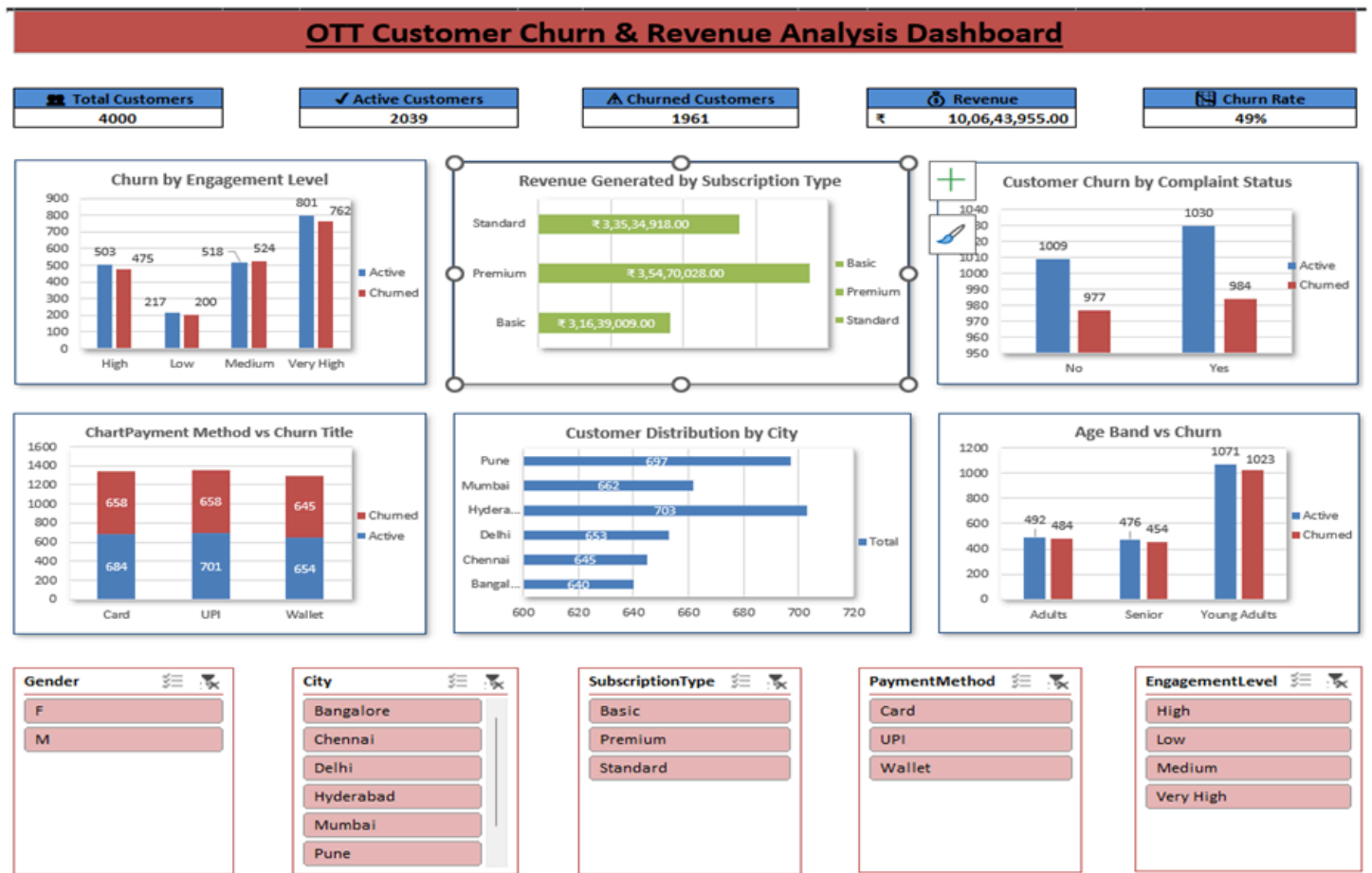


Figure 1: Final OTT Customer Churn Analysis Dashboard

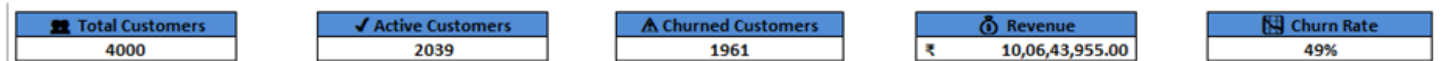


Figure 2: Final OTT Customer Churn Analysis KPI's

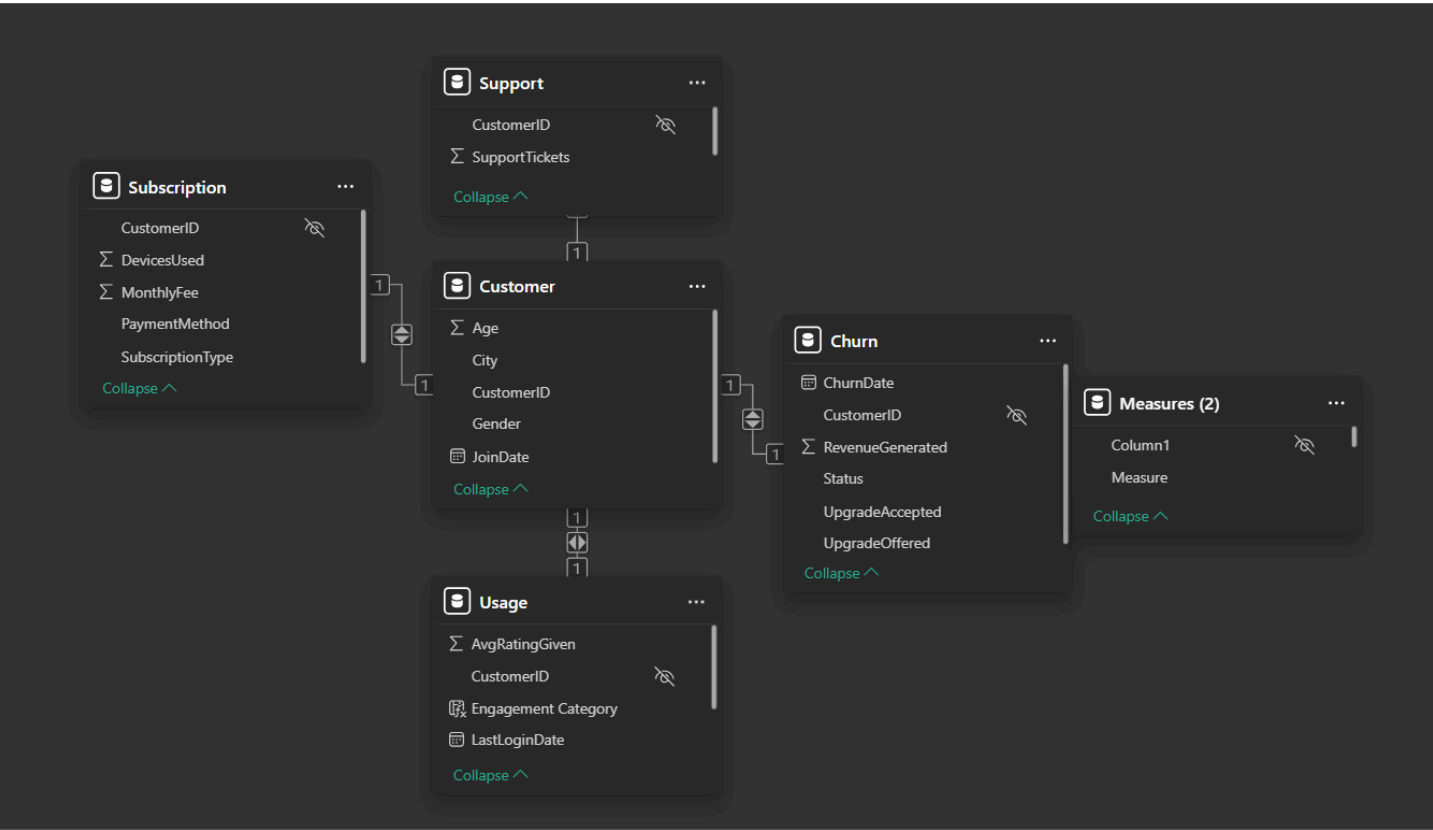


Figure 2: Final OTT Customer Churn Analysis Data Model

Approval

Prepared By : Abhijeet Kedar

Reviewed By :

Approved By :